



Project 2: Exploratory Data Analysis (EDA)

Goal:

Analyze a dataset to understand patterns, trends, and distributions.

Key Requirements:

- Calculate basic statistics (mean, median, count)
- Identify trends and outliers
- Summarize key observations

Key Skills:

Data analysis, descriptive statistics, analytical thinking

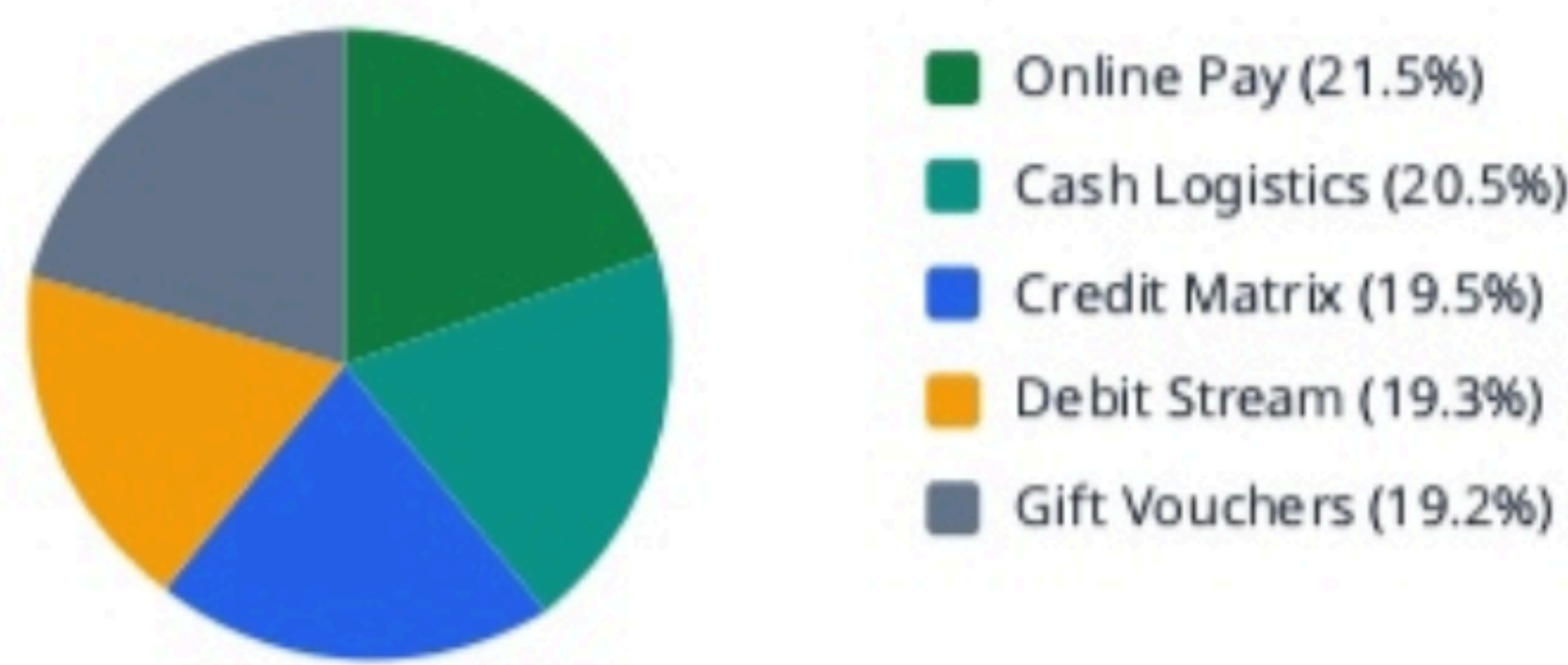
fx =COUNTA('Cleaned_Data_Analytics'!A2:A1201)

Sales Transaction Performance Dashboard

This section summarizes the high-level metrics evaluated directly from your uploaded **Cleaned_Data_Analytics_Dataset-2**. The system records an extensive array of transactions mapping operations securely.

TOTAL SALES ORDERS (N)	AVERAGE ORDER VALUE	MAX TRANSACTION VALUE
1,200	\$1,053.97	\$3,456.40

Distribution of System Payment Preferences (%)




```
fx  =MEDIAN('Cleaned_Data_Analytics'!N2:N1201)
```

Five-Number Mathematical Summary Matrix

The continuous metrics column (**TotalPrice**) was evaluated using standard percentile cuts to compute the baseline geometric spread parameters.

Statistical Parameter Name	Calculated Value (\$)	Excel Operational Formula Block
Minimum (The Data Floor)	\$11.39	=MIN(TotalPrice_Range)
25th Percentile (Q1 Threshold)	\$410.52	=PERCENTILE.INC(TotalPrice_Range, 0.25)
Median (50th Percentile / Center)	\$823.62	=MEDIAN(TotalPrice_Range)
75th Percentile (Q3 Threshold)	\$1,578.48	=PERCENTILE.INC(TotalPrice_Range, 0.75)
Maximum (The Data Ceiling)	\$3,456.40	=MAX(TotalPrice_Range)
Arithmetic Average Value (Mean)	\$1,053.97	=AVERAGE(TotalPrice_Range)

Volume Stream Mapped across Inventory Product Lines



Production Order Extract (Audit Ledger)

A validation trace snippet demonstrating structural alignment matching values from row index parameters.

OrderID	Date Index	Product	Qty	Payment	Total Price	System Status
ORD200000	04-01-2023	Monitor	5	Debit Card	\$2,853.10	Shipped
ORD200001	23-08-2024	Phone	2	Online	\$302.70	Shipped
ORD200002	27-02-2024	Tablet	5	Credit Card	\$2,753.40	Cancelled
ORD200003	15-10-2023	Chair	1	Debit Card	\$273.19	Returned
ORD200004	08-05-2025	Printer	4	Online	\$2,504.04	Delivered
ORD200005	12-09-2023	Laptop	3	Cash	\$1,432.10	Shipped
ORD200006	19-11-2024	Desk	1	Gift Card	\$412.50	Delivered